Digital Assignment-4

Name: Kanish Kumar S

Reg.no: 23MIS0141

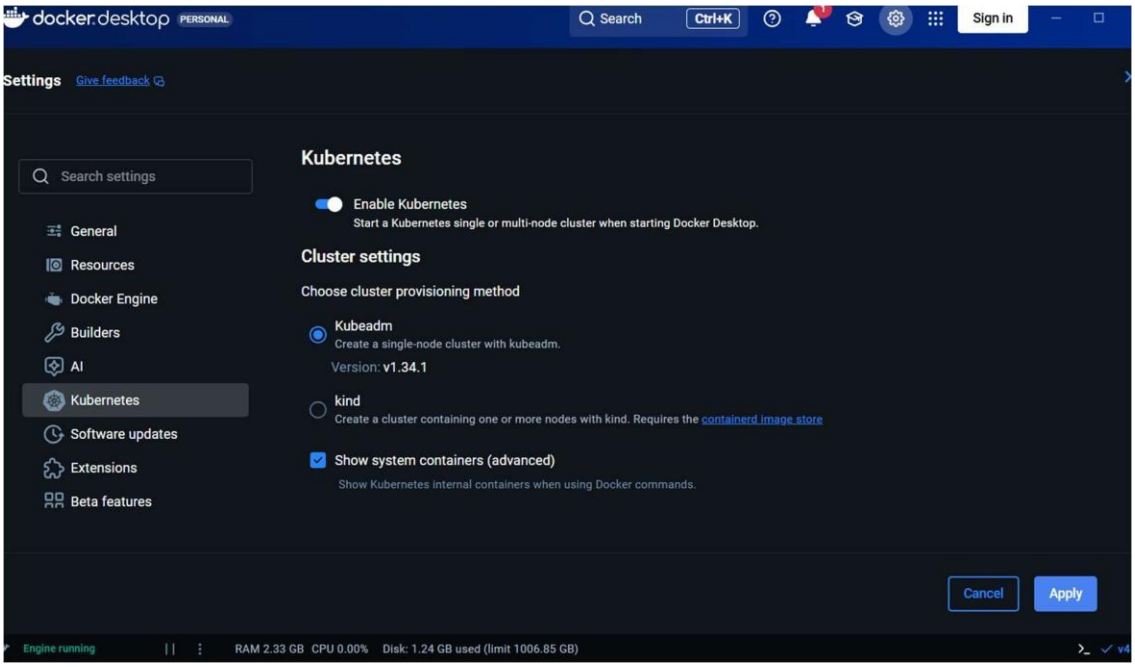
Course: Agile Development process and Devops lab

Faculty: Prof. Bala ganesh

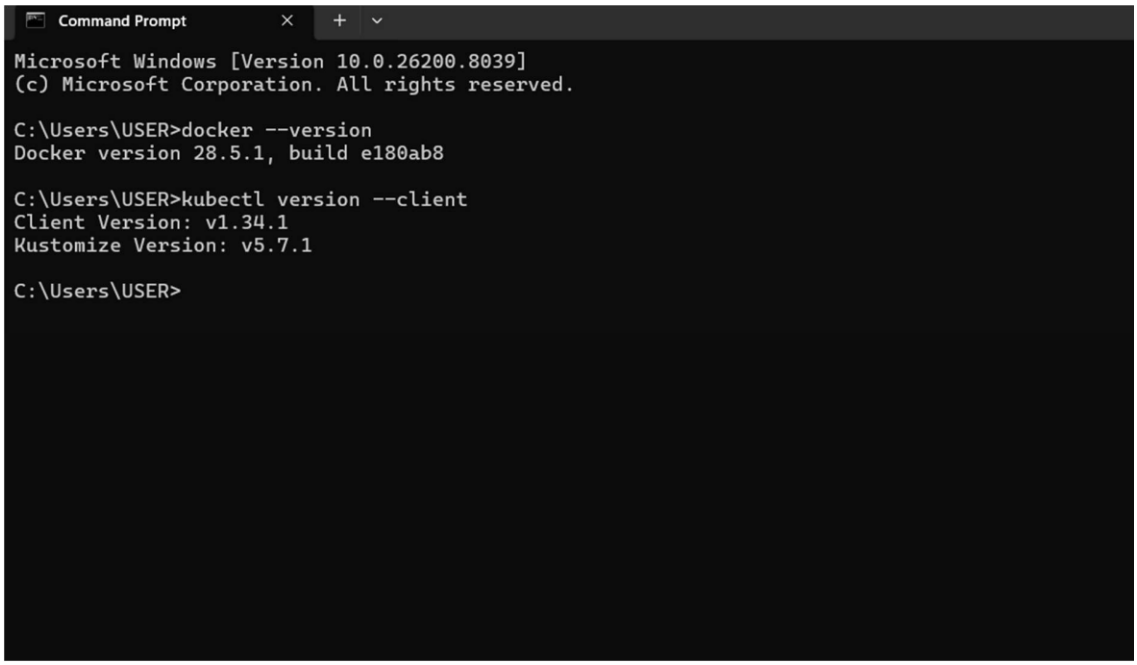
Step1:

Install Docker Desktop, enable Kubernetes, enable three plugins (Docker, Docker Pipeline, Kubernetes CLI) and install them

Enable Kubernetes:

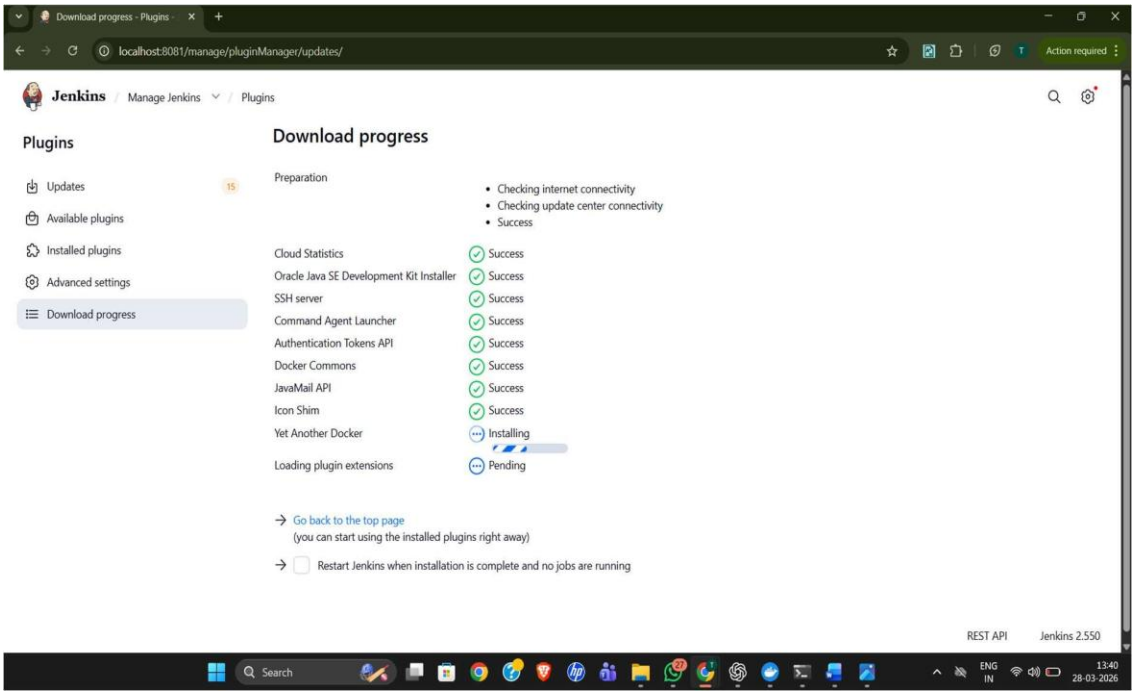


Verify Installation:

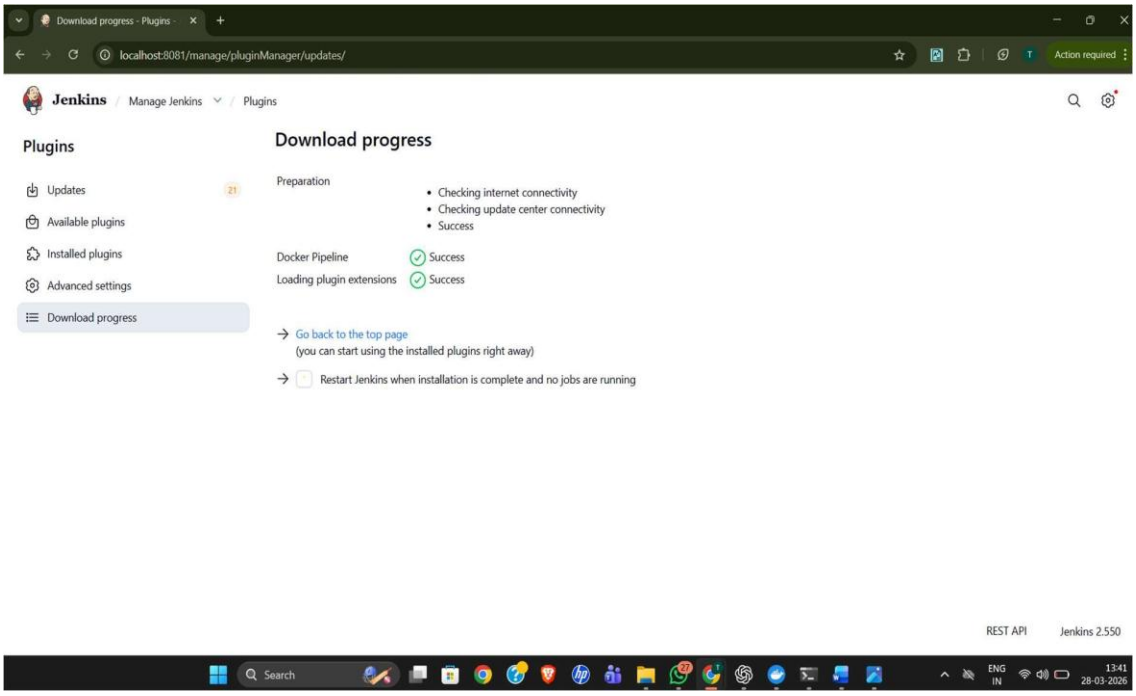


Install Plugins:

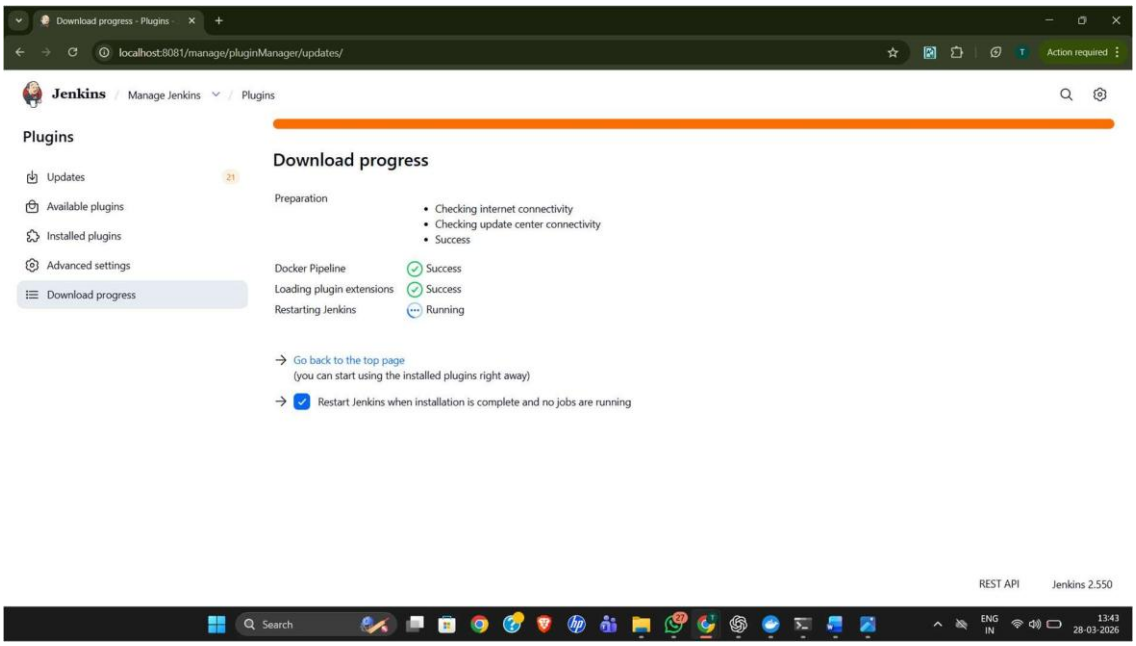
Docker Plugin :



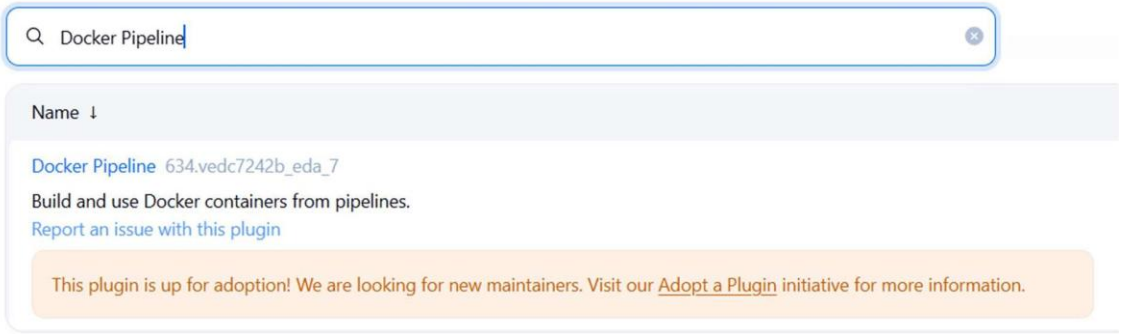
Docker Pipeline



Kubernetes CLI Plugin:



Verify Installation:

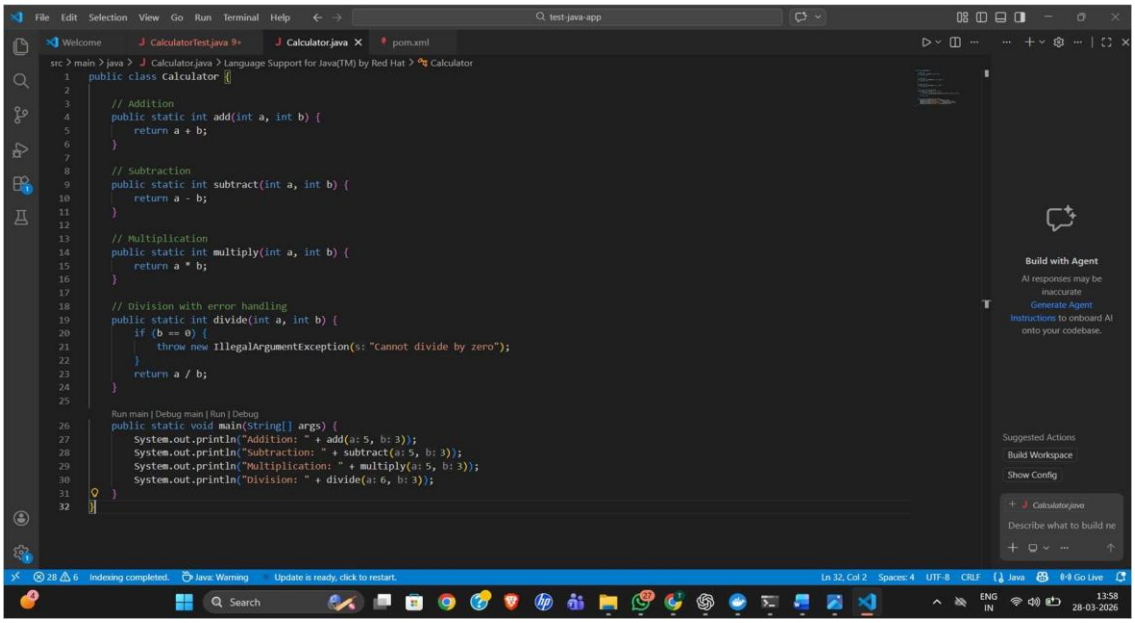


Step 2 - Create Simple Java Application (in local machine) The folder structure will be,

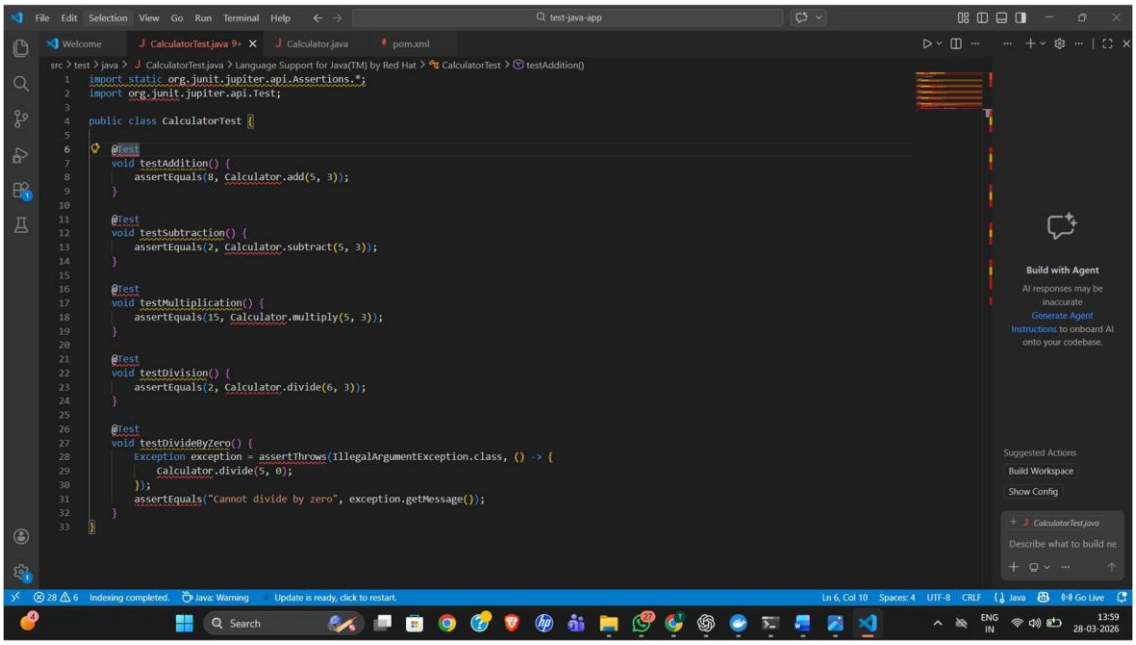
test-java-app

```
│
├── src
│   ├── main
│   │   └── java
│   │       └── Calculator.java
│   │
│   └── test
│       └── java
│           └── CalculatorTest.java
│
└── pom.xml
```

Main Java Code (Calculator.java)

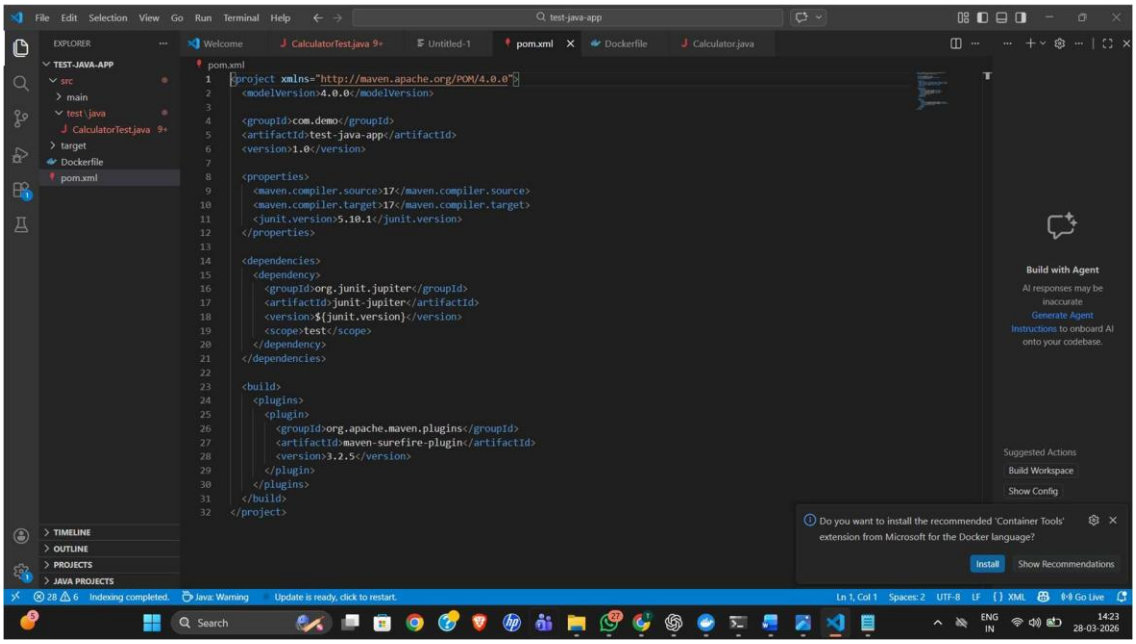


JUnit Test Code (CalculatorTest.java)



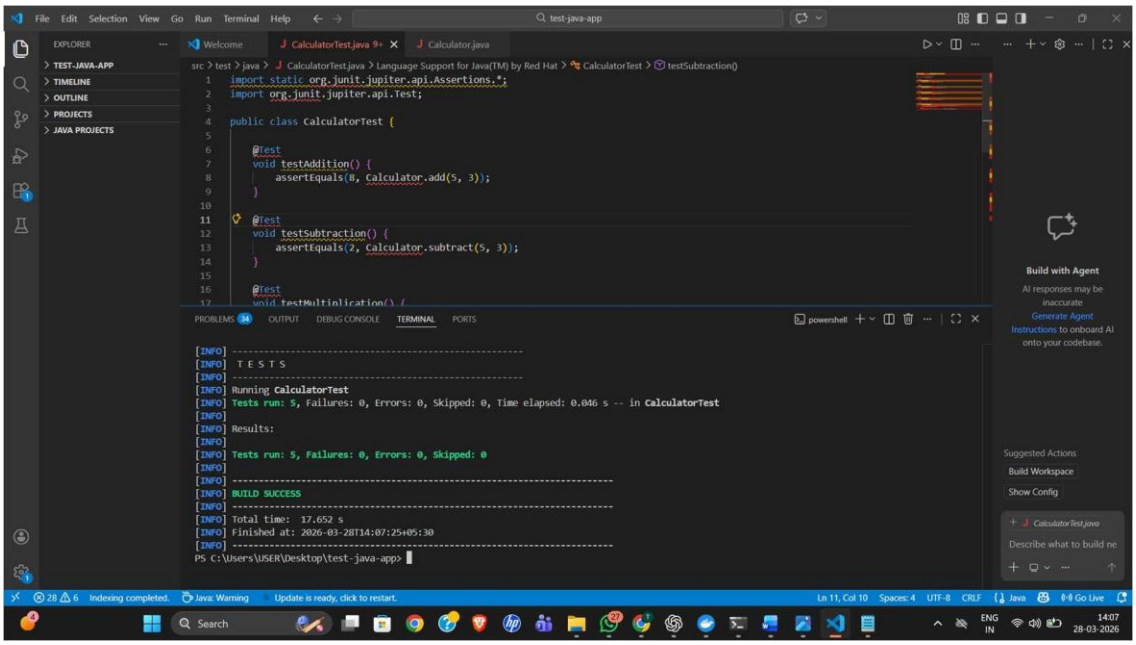
```java
import static org.junit.jupiter.api.Assertions.*;
import org.junit.jupiter.api.Test;

public class CalculatorTest {

    @Test
    void testAddition() {
        assertEquals(8, Calculator.add(5, 3));
    }

    @Test
    void testSubtraction() {
        assertEquals(2, Calculator.subtract(5, 3));
    }

    @Test
    void testMultiplication() {
        assertEquals(15, Calculator.multiply(5, 3));
    }

    @Test
    void testDivision() {
        assertEquals(2, Calculator.divide(6, 3));
    }

    @Test
    void testDivideByZero() {
        Exception exception = assertThrows(IllegalArgumentException.class, () -> {
            Calculator.divide(5, 0);
        });
        assertEquals("Cannot divide by zero", exception.getMessage());
    }
}
```

pom.xml:

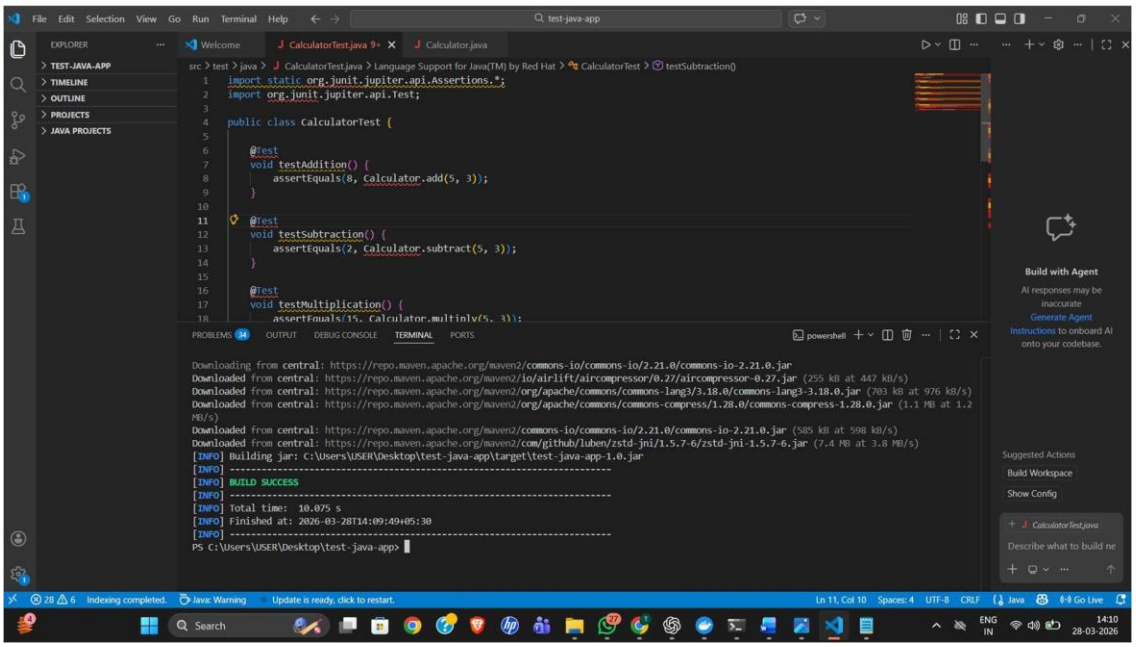


```xml
<project xmlns="http://maven.apache.org/POM/4.0.0">
  <modelVersion>4.0.0</modelVersion>

  <groupId>com.demo</groupId>
  <artifactId>test-java-app</artifactId>
  <version>1.0</version>

  <properties>
    <maven.compiler.source>17</maven.compiler.source>
    <maven.compiler.target>17</maven.compiler.target>
    <junit.version>5.10.1</junit.version>
  </properties>

  <dependencies>
    <dependency>
      <groupId>org.junit.jupiter</groupId>
      <artifactId>junit-jupiter</artifactId>
      <version>${junit.version}</version>
      <scope>test</scope>
    </dependency>
  </dependencies>

  <build>
    <plugins>
      <plugin>
        <groupId>org.apache.maven.plugins</groupId>
        <artifactId>maven-surefire-plugin</artifactId>
        <version>3.2.5</version>
      </plugin>
    </plugins>
  </build>
</project>
```

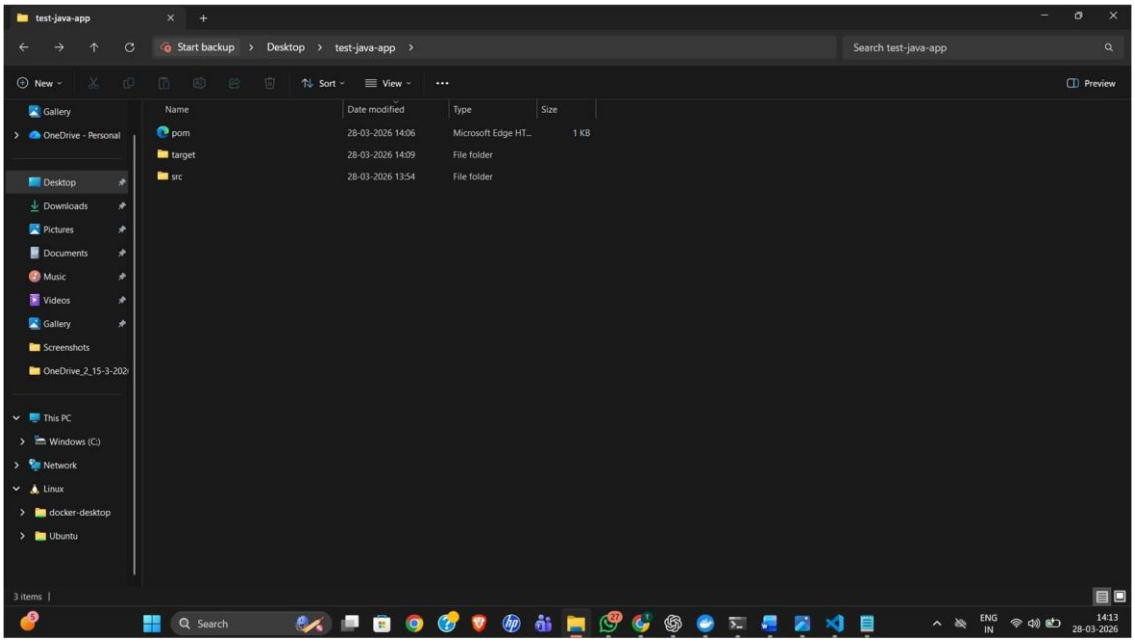
Run mvn clean test:

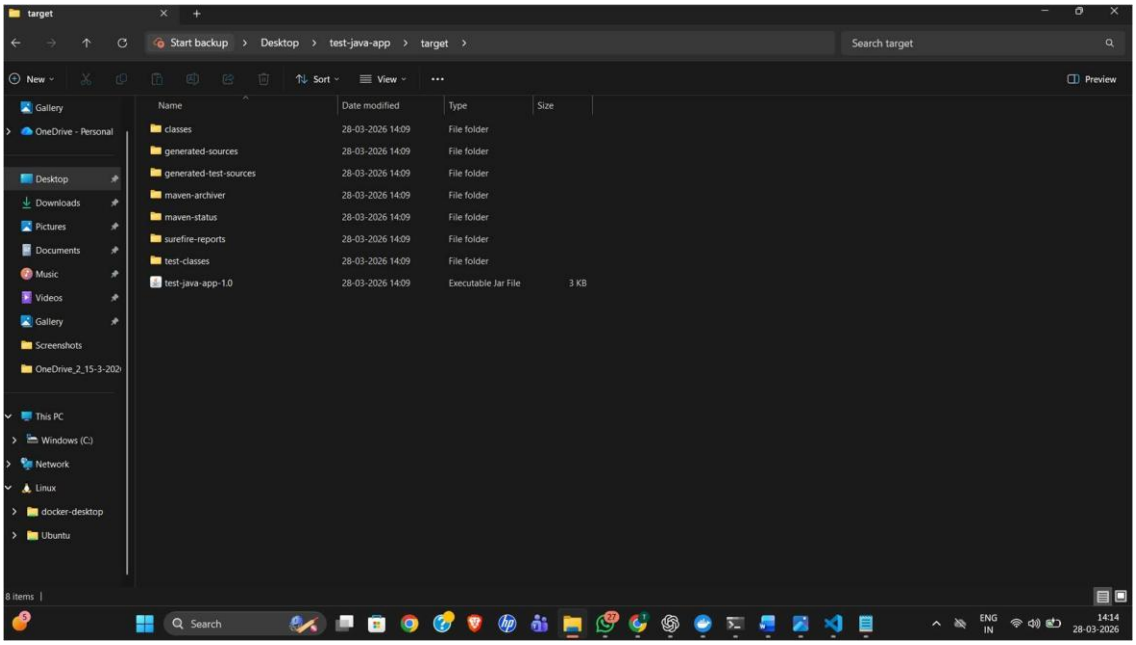


Run mvn clean package:



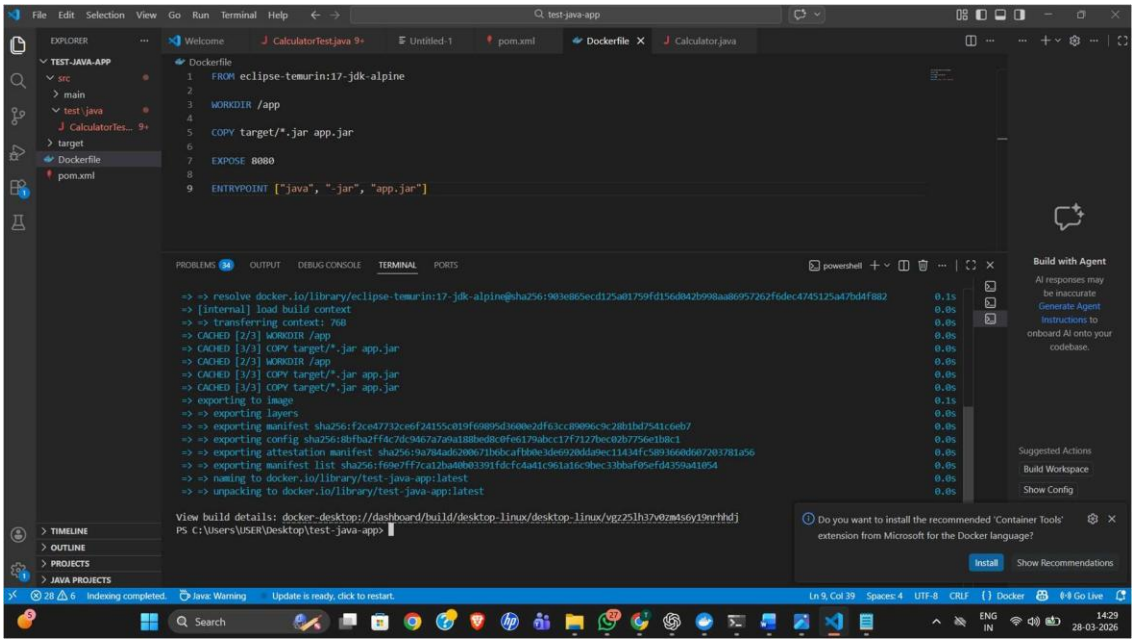
It generates: target/ test-java-app-1.0.jar



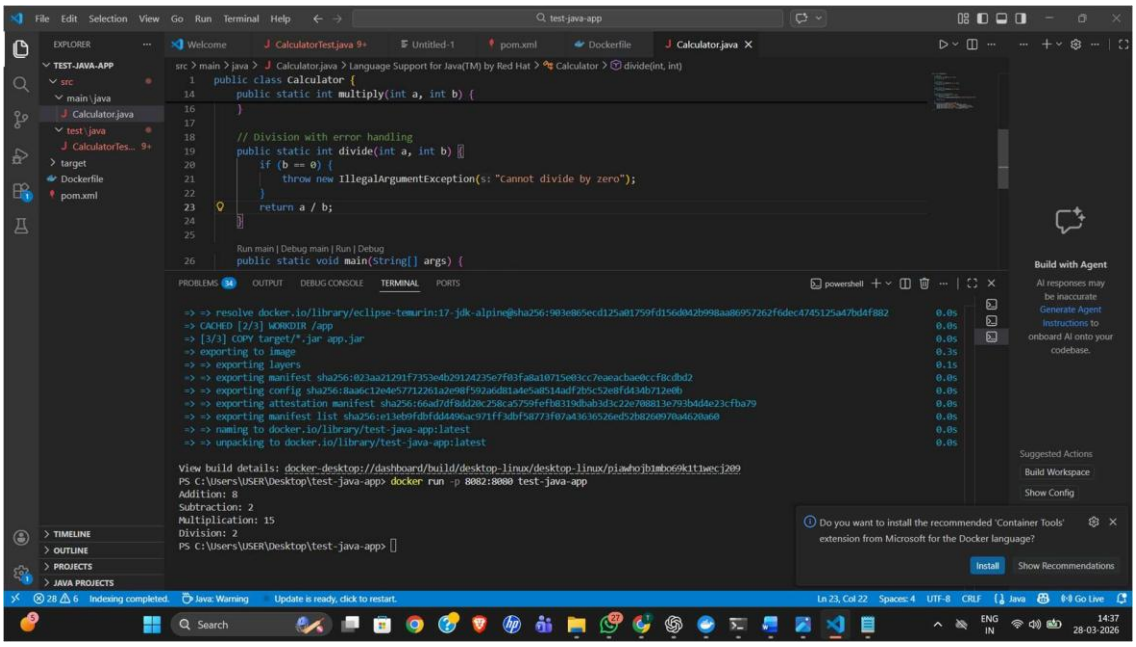


Step 4 - Create Dockerfile Dockerfile

Create Docker file and run docker build -t test-java-app :

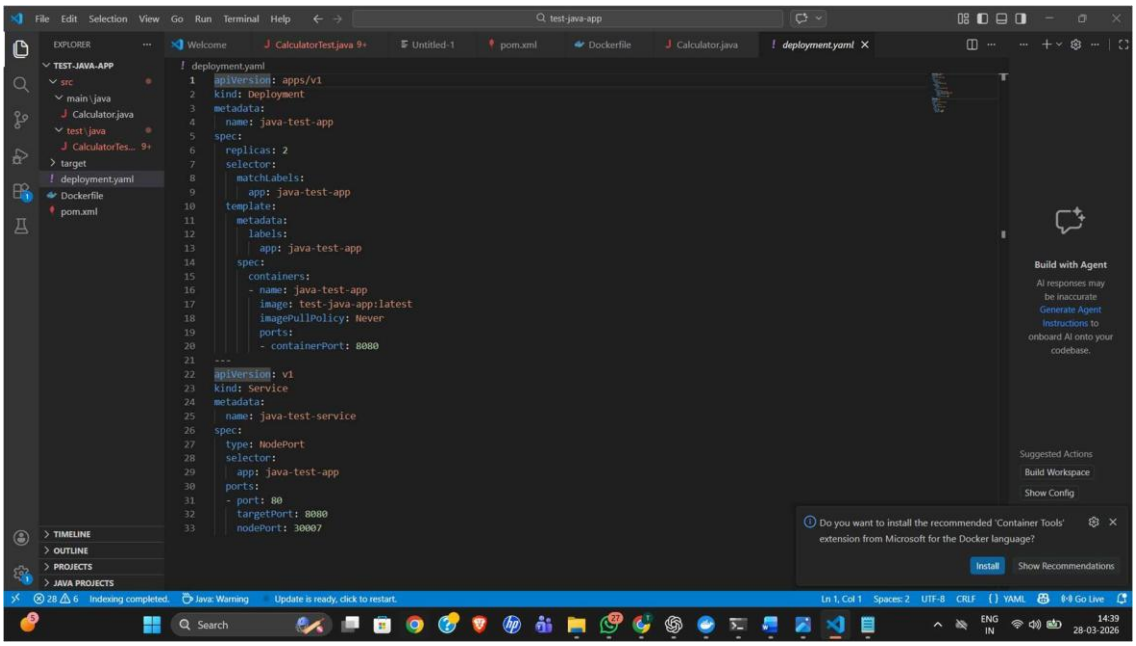


docker run -p 8082:8080 test-java-app:



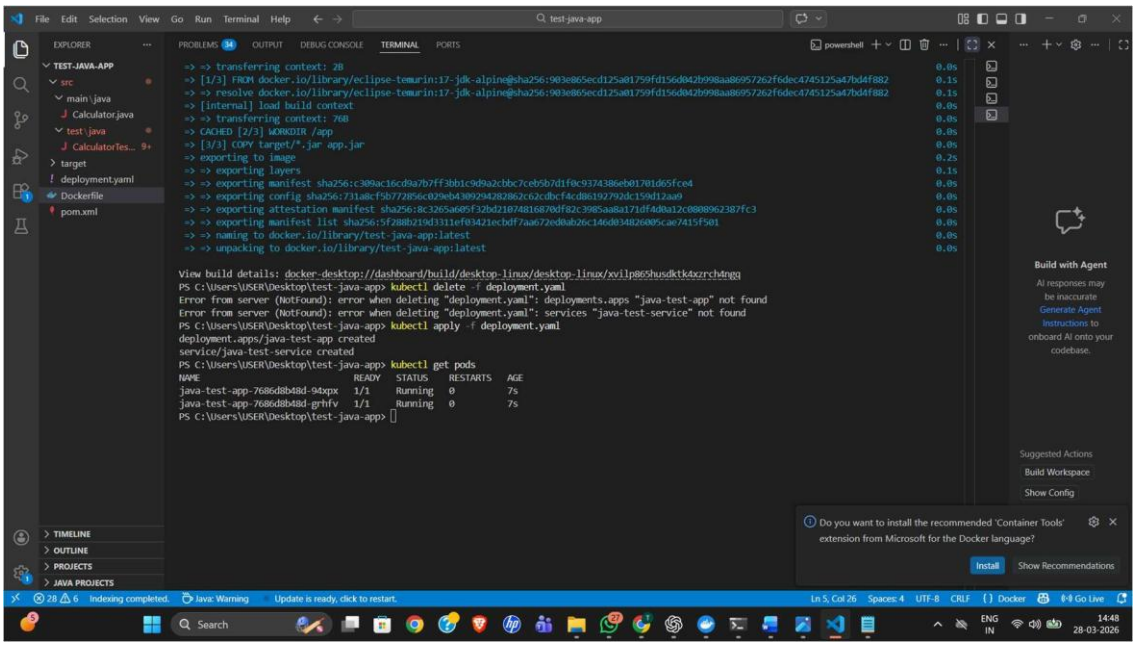
Step 5 - Create Kubernetes Deployment File

deployment.yaml:

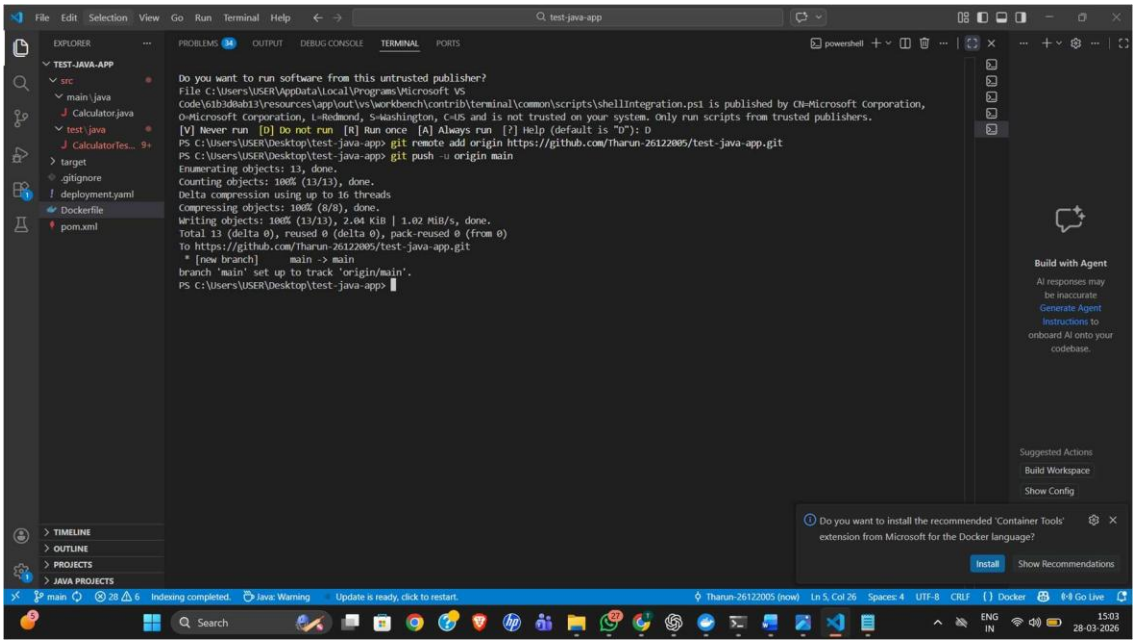


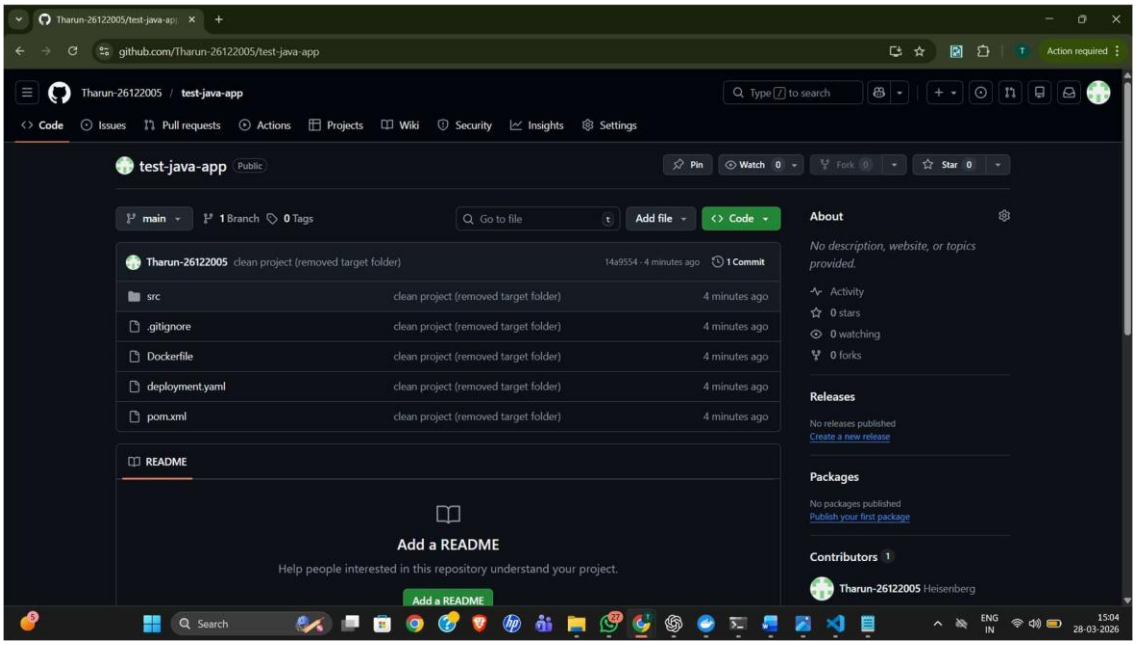
kubectl apply -f deployment.yaml:

kubectl get pods:

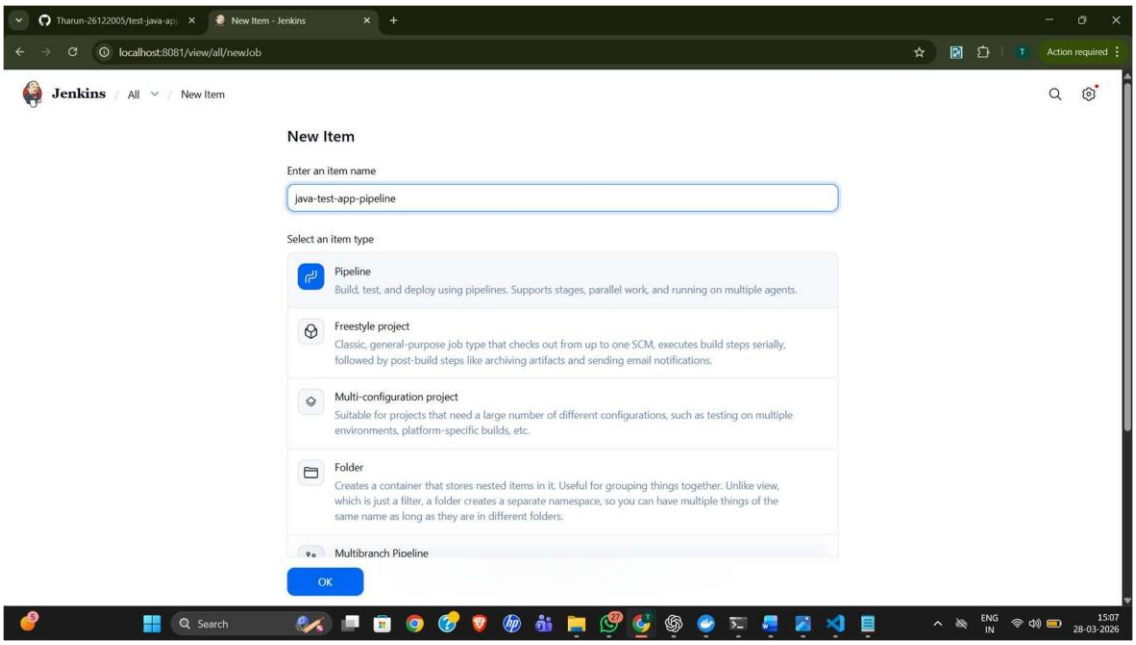


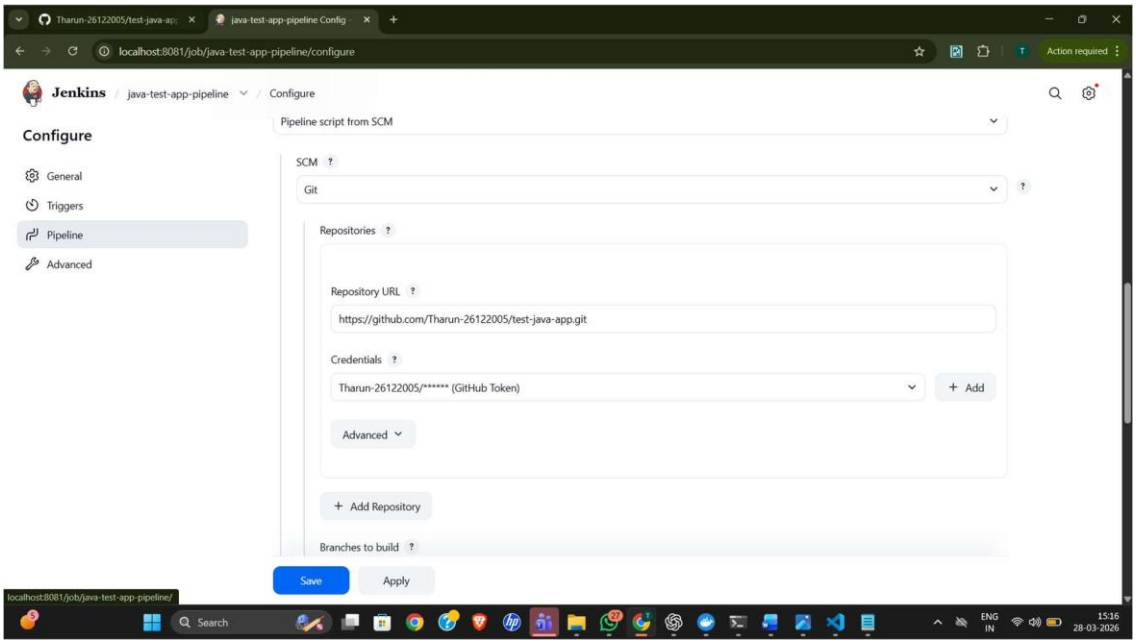
Step 6 - Push Code to GitHub



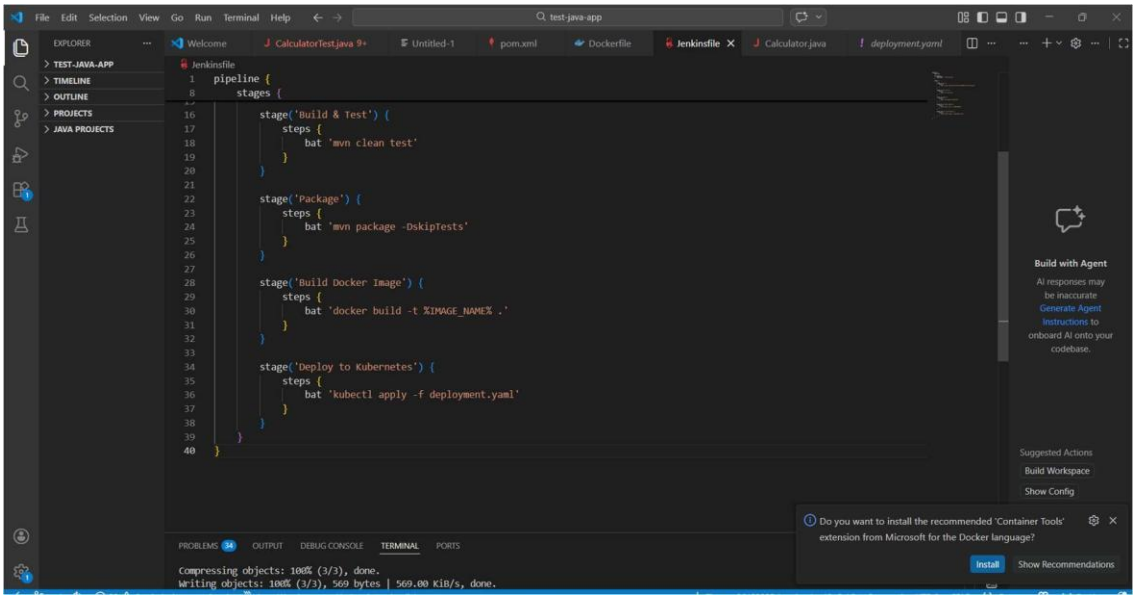


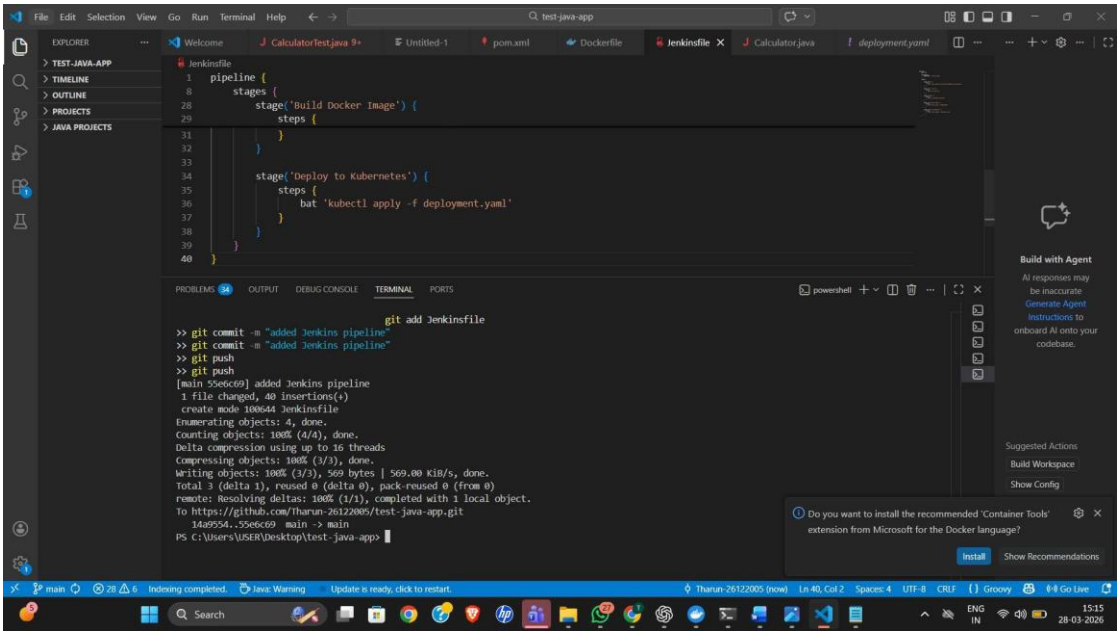
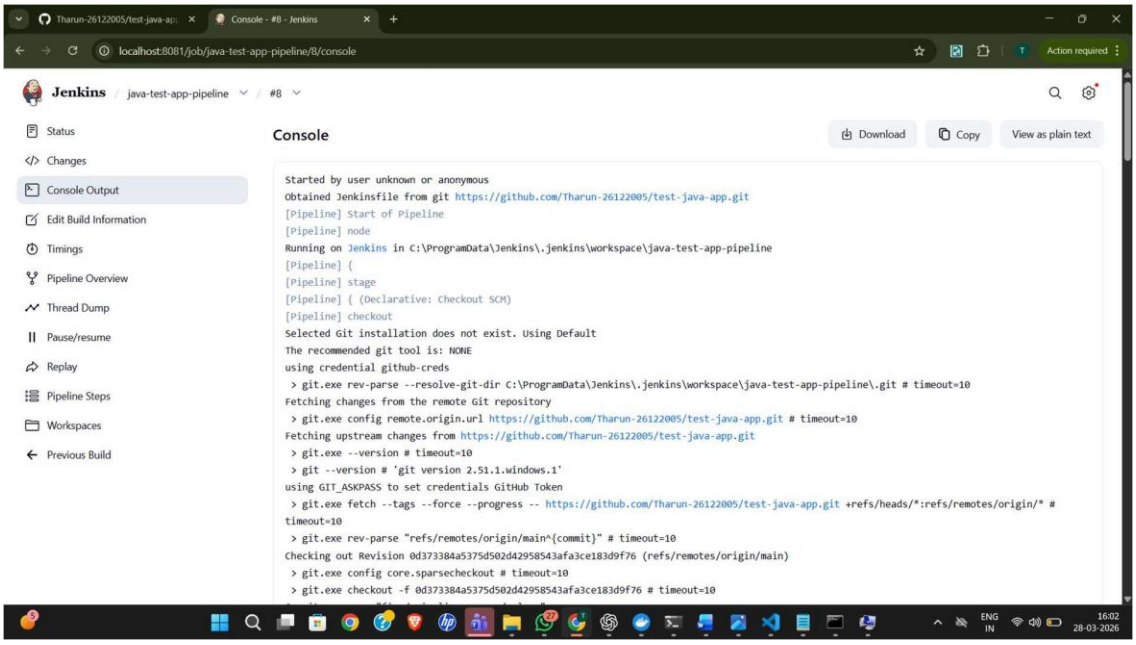
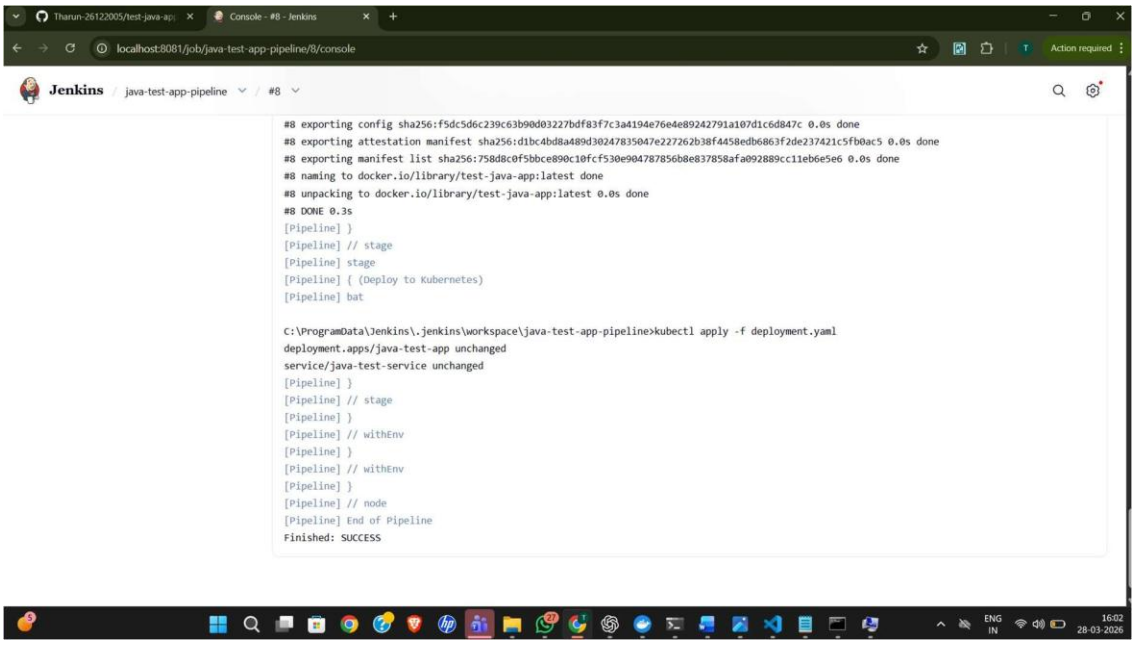
Step 7 - Create Jenkins Pipeline Job





Jenkinsfile:





Run Pipeline: